

Physics 1 (J)

Points: 18/20

Time: 21:12

✓ **Correct** 1/1 Points

1

If two points on an equipotential surface are labeled A and B, what is the potential difference (ΔV) between them? *



- ☒ $\Delta V = 0$
- ☐ $\Delta V > 0$
- ☐ $\Delta V < 0$
- ☐ ΔV depends on the charge of the particle.

✓ **Correct** 1/1 Points

2

In a spherical capacitor, where is the electric field strongest? *



- ☒ At the inner sphere
- ☐ At the outer sphere
- ☐ At the midpoint between the spheres

☐ Electric field strength is the same everywhere.

✓ **Correct** 1/1 Points

3

If the displacement vector ds is perpendicular to the electric field vector E , what is the work done according to the equation?

* 

- ☐ Maximum positive work
- ☐ Maximum negative work
- ☒ Zero work
- ☐ Work depends on the charge at point i .

✓ **Correct** 1/1 Points

4

At an infinite distance from an electric dipole, the electric potential approaches:

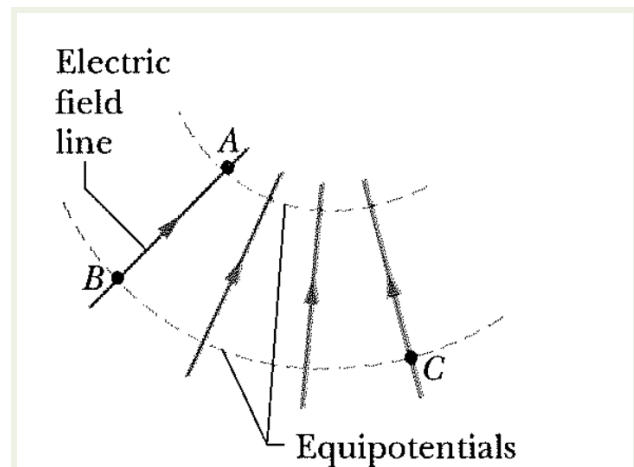
* 

- ☒ Zero
- ☐ A finite non-zero value
- ☐ Infinity
- ☐ Cannot be determined.

✓ **Correct** 2/2 Points

5

When an electron moves from A to B along an electric field line in the adjacent figure, the electric field does 3.94×10^{-19} J of work on it. What are the electric potential differences $V_C - V_A$? *



- ☐ 0 V
- ☐ 1.45 V
- ☒ 2.45 V
- ☐ - 2.45 V

✓ **Correct** 2/2 Points

6

The ammonia molecule $[\text{NH}_3]$ has a permanent electric dipole moment equal to 1.47 D, where $1\text{D} = 1\text{debye unit} = 3.34 \times 10^{-30}$ C.m. Calculate the electric potential due to an ammonia molecule at a point 52.0 nm away along the axis of the dipole. (Set $V=0$ at infinity.)

*

- ☐ 1.634×10^{-6} Volt
- ☒ 16.34×10^{-6} Volt
- ☐ $16.34 \times 10^{(6)}$ Volt

☐ $1.634 \times 10^{(6)}$ Volt

✓ **Correct** 2/2 Points

7

You have two flat metal plates, each of area 1 m^2 , with which to construct a parallel-plate capacitor. If the capacitance of the device is to be 2 F , what must be the separation between the plates? * 

☐ $8.854 \times 10^{(-12)}$ m

☐ $8.854 \times 10^{(12)}$ m

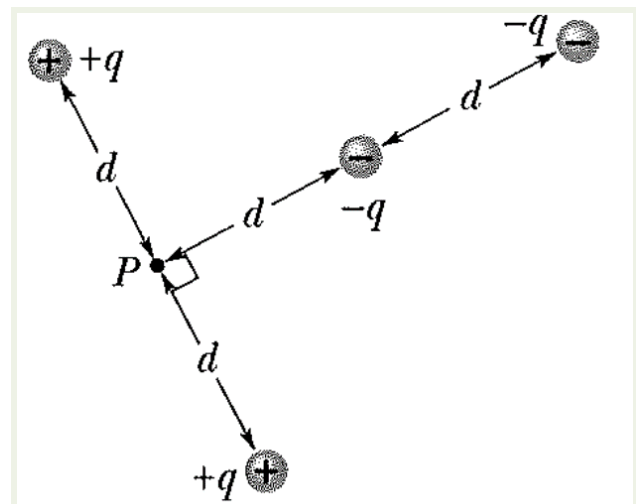
☒ $4.427 \times 10^{(-12)}$ m

☐ $4.427 \times 10^{(12)}$ m

✓ **Correct** 2/2 Points

8

In the adjacent Figure, what is the net electric potential at point P due to the four particles if $V=0$ at infinity, $q=5 \text{ fC}$, and $d=4 \text{ cm}$? * 



☐ $562.5 \times 10^{(6)}$ Volt

☐ $56.25 \times 10^{(-6)}$ Volt

☐ $5.625 \times 10^{(-6)}$ Volt

☒ $562.5 \times 10^{-6} \text{ Volt}$

✓ **Correct** 1/1 Points

9

In a spherical capacitor, what happens to the capacitance if the separation between the spheres is doubled? * 

- ☐ It doubles
- ☒ It halves
- ☐ It quadruples
- ☐ It remains the same.

✓ **Correct** 2/2 Points

10

The electric potential V in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 150x^2$, where x (in meters) is the perpendicular distance from plate 1. At $x = 1.3 \text{ cm}$, what is the magnitude of the electric field * 

- ☒ 3.9 V/m
- ☐ 39 V/m
- ☐ 39 V/cm
- ☐ 3.9 V/cm

✓ **Correct** 2/2 Points

11

What is the escape speed for an electron initially at rest on the surface of a sphere with a radius of 1.0 cm and a uniformly distributed charge of 1.6×10^{-15} C? That is, what initial speed must the electron have in order to reach an infinite distance from the sphere and have zero kinetic energy when it gets there?

* 

- ☐ 22 m/s
- ☒ 22 Km/s
- ☐ 22 Km/m
- ☐ 22 Km/h

✗ **Incorrect** 0/1 Points

12

If a positive charge moves in the direction of the electric field, the change in electric potential *



- ☐ Positive
- ☒ Negative
- ☐ Zero
- ☐ Cannot be determined.

✓ **Correct** 1/1 Points

13

In a parallel combination of capacitors, how are the charges on each capacitor related to the total charge (total Q total)?

* 

- ☐ The charge on each capacitor is equal to total Q total.
- ☐ The charge on each capacitor is inversely proportional to its capacitance.
- ☒ The charge on each capacitor is directly proportional to its capacitance.
- ☐ The charge on each capacitor is zero.

✗ **Incorrect** 0/1 Points

14

If the charge on a capacitor is doubled while the voltage remains constant, how does the capacitance change? * 

- ☒ It doubles
- ☐ It halves
- ☐ It remains the same
- ☐ It becomes four times.

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